Wolfgang William Schmaltz

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Education

University of California, Berkeley Aug. 2012 – May 2018
Doctor of Philosophy in Mathematics

University of Chicago Sep. 2007 – Jun. 2011
Bachelor of Science in Mathematics

Research Positions

Polyfold Theorist Mar. 2020 – Dec. 2024
Faculty of Mathematics, Ruhr-Universität Bochum *Bochum, Germany*

Polyfold Theorist Jan. 2018 – Feb. 2020
Mathematics Institute, Justus-Liebig University *Gießen, Germany*

Papers

Pseudocycle Gromov–Witten invariants are a strict subset of polyfold Gromov–Witten invariants Aug. 2023
W. Schmaltz *arXiv:2308.14204*

Non-fillability of overtwisted contact manifolds via polyfolds Nov. 2020
W. Schmaltz, S. Suhr, K. Zehmisch *arXiv:2011.02249*

The Gromov–Witten axioms for symplectic manifolds via polyfold theory Dec. 2019
W. Schmaltz *arXiv:1912.13374*

Naturality of polyfold invariants and pulling back abstract perturbations Dec. 2019
W. Schmaltz *arXiv:1912.13370*

The Steenrod problem for orbifolds and polyfold invariants as intersection numbers Apr. 2019
W. Schmaltz *arXiv:1904.02186*

Selected Invited Talks

Orbifold Intersection Theory and Polyfold Gromov–Witten Invariants Nov. 28, 2023
Heidelberg Geometry Seminar *Heidelberg, Germany*

The Steenrod problem for orbifolds and polyfold Gromov–Witten invariants Dec. 10, 2020
Oberseminar Dynamische Systeme *Bochum, Germany*

Gromov–Witten Axioms for Symplectic Manifolds via Polyfold Theory Oct. 24, 2018
SISSA *Trieste, Italy*

Naturality of Polyfold Invariants May 2, 2018
Mathematical Sciences Research Institute *Berkeley, California*

The Polyfold Gromov–Witten Invariants & Gromov–Witten Axioms for Symplectic Manifolds via Polyfold Theory Apr. 25, 2018
Berkeley Topology Seminar *Berkeley, California*

Projects

LLMs from the Mathematical Viewpoint www.wolfgangschmaltz.com/llm_mathematics

- A mathematics first presentation of the fundamental theory of Large Language Models, designed specifically for an audience of mathematicians. Using mathematical terminology and formula, explains the definitions and constructions of tokenizers, causally masked attention, the transformer architecture, and a language model itself. Basic implementations of these constructions in PyTorch are also included to act as a bridge between the abstract mathematical formulations and the implementation in code.

QLoRA Fine-Tuning to Generate Stories www.wolfgangschmaltz.com/stories

- A QLoRA adapted small language model which generates stories based upon discrete user-specified criteria: theme, topic, word count, and complexity. Based on a small (35 million parameters) language model, and trained using supervised fine-tuning, QLoRA, and quantization. A user interface for interacting with the model, including a text streaming effect, was built using Streamlit. Accessible as a webapp, hosted on Hugging Face, Streamlit, and my personal website.

Skills

Programming Languages: Python (excellent), SQL (excellent), Java, Bash
Foreign Languages: English (native), German (C1), Turkish (A2)